A prospective double-blind randomized placebo-controlled study of the effect of saffron (Crocus sativus Linn.) on semen parameters and seminal plas...

Male factor infertility is a multifactorial disorder that affects a significant percentage of infertile couples; however, many of them remained untreated. In recent years, considerable numbers of infertile men have sought 'herbal remedies' as an effective treatment. Among 'herbal remedies', saffron is recommended for male infertility in our community. The effect of saffron was evaluated compared with placebo for the treatment of idiopathic male factor infertility. The study included 260 infertile men with idiopathic oligoasthenoteratozoospermia (OAT) who were randomized to saffron 60 mg/day (130, group 1) or a similar regimen of placebo (130, group 2) for 26 weeks. The two groups were compared for changes in semen parameters and total seminal plasma antioxidant capacity. Saffron administration did not result in beneficial effects. At the end of the study no statistically significant improvements were observed in either group in any of the studied semen parameters (sperm density, morphology and motility) (all $p = 0.1$). At the end of the trial, patients in group 1 had a mean motility of $25.7 \pm 2.4\%$, which was not statistically different from the mean of $24.9 \pm 2.8\%$ in the placebo group ($p = 0.1$). Normal sperm morphology was $18.7 \pm 4.7\%$ and $18.4 \pm 4.3\%$, in groups 1 and 2, respectively ($p = 0.1$). Patients treated with saffron and placebo had a mean sperm density of $20.5 \pm 4.6\%$ and $21.4 \pm 4.6\%$ per mL, respectively ($p = 0.1$). Saffron administration did not improve total seminal plasma antioxidant capacity, compared with baseline ($p = 0.1$) and placebo subjects ($p = 0.1$). Based on Pearson correlations, each semen parameter did not correlate significantly with treatment duration, including sperm density ($r = 0.146$, $p = 0.13$), percent of motile sperm ($r = 0.145$, $p = 0.15$) and percent of sperm with normal morphology ($r = 0.125$, $p = 0.30$). Saffron does not statistically significantly improve semen parameters in infertile men with idiopathic OAT. If medical professionals want to prescribe herbal remedies for male infertility, previous rigorous scientific investigations, documenting their safety and efficacy are required.